

ACSM 2026 Position Stand
Resistance Training Prescription Overview

American College of Sports Medicine
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Core Finding
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Resistance training improves muscle strength, hypertrophy, power, endurance, contraction velocity, and physical function in healthy adults. However, only a few prescription variables significantly enhance adaptations beyond the basic effect of doing RT versus doing nothing.

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What Truly Matters for Strength
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Voluntary strength is enhanced by: lifting heavier loads (â%ºŸ80% one-repetition maximum, with a dose-response relationship), training through a complete range of motion, performing 2â€”3 sets per exercise, placing the exercise at the beginning of training sessions, and training â%ºŸ2 sessions per week.

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What Truly Matters for Hypertrophy
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Muscle hypertrophy is enhanced by: higher volumes (â%ºŸ10 sets per muscle group per week, with a dose-response relationship) and eccentric overload. Notably, load magnitude does not consistently impact hypertrophyâ€”low loads (30% 1RM) to high loads (100% 1RM) produce similar gains when effort is near-fatiguing.

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What Truly Matters for Power
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Muscle power is enhanced by: moderate loads (30%â€”70% 1RM), low-to-moderate volume (â%ºº24 repetitions Å—sets), Olympic-style weightlifting, and power RT (intentionally performing the concentric phase at maximal volitional speed). Power RT also enhances physical function outcomes.

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Variables That Do NOT Consistently Matter
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Training to momentary muscle fatigue is not necessary for strength or hypertrophy gains and may increase injury risk, particularly in older adults. Equipment type (machines vs free weights), exercise complexity (single-joint vs multi-joint), set structure (cluster, drop sets, complex, contrast), time under tension, blood flow restriction, and periodization do not consistently impact training outcomes when volume and effort are equated.

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Physical Function: The Lower Extremity Connection
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RT significantly improves gait speed, balance, chair stand test performance, timed up-and-go, and multicomponent functional assessments. These are clinically relevant outcomes for aging populations and rehabilitation settings. The evidence confirms that lower extremity strength training is a primary driver of functional independence.

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